General T3 EFT1 one-loop RGEs

Native RGBeta expressions: 16 pi squared times dX / dln(mu)

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1 A__+0

Additional quartics: 5; additional bilinears: 1; additional Yukawa: y1Conj
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

23 beta functions

```
beta_gY =
(47*gY^3)/6

beta_g2 =
(-5*g2^3)/2

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
4*lambdaS1*mS1Sq + 6*lambda12*mS2Sq + 12*muT3GeneralS1Sq*Bar[lambdaT3GeneralQ001300I1] + 12*lambdaT3GeneralQ001300I1
    *Bar[muT3GeneralS1Sq]

beta_mS2Sq =
2*lambda12*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq + (8*lambdaS2 + 4*lambdaS2Adj)*mS2Sq + 4*muT3GeneralS1Sq*Bar[
    lambdaT3GeneralQ000211I1] + 4*lambdaT3GeneralQ000211I1*Bar[muT3GeneralS1Sq]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 2*lambdaH1^2 + 6*lambdaH2
    ^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 8*
    lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[
    Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[
    Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
6*lambda12^2 + 4*lambdaH1^2 + 10*lambdaS1^2 + 12*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1] + 288*
    lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000400I1] + 108*lambdaT3GeneralQ001300I1*Bar[
    lambdaT3GeneralQ001300I1] + 8*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1]

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH2^2 + gY^2*(8*g2^2 - 12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 +
    8*lambdaS2*lambdaS2Adj + 16*lambdaS2Adj^2 + 8*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1]

beta_lambdaH1 =
(-9*g2^2*lambdaH1)/2 - (3*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambda12*lambdaH2 + 4*lambdaH1*
    lambdaS1 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 12*
    lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ110200I1] + 4*lambdaT3GeneralQ110200I1*(3*Bar[
    lambdaT3GeneralQ001300I1] + 4*Bar[lambdaT3GeneralQ110200I1]) + lambdaH1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + 2*lambda12*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*
    lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ110200I1*Bar[
    lambdaT3GeneralQ000211I1] + 4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 4*
    lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH2*(4*lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] +
    2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 4*lambda12*lambdaS1 + 8*lambda12*lambdaS2
     + 4*lambda12*lambdaS2Adj + 4*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ000211I1
```

```
    ] + 4*lambdaT3GeneralQ000211I1*(4*Bar[lambdaT3GeneralQ000211I1] + 3*Bar[lambdaT3GeneralQ001300I1]) + 4*
    lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + 4*Bar[lambdaT3GeneralQ021010I1]*(Bar[lambdaT3GeneralQ000211I1] + 2*Bar[
    lambdaT3GeneralQ110200I1]) + lambdaT3*(4*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
    6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
gY^2*(4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*lambdaH2Adj + 2*
    lambdaH2Adj*lambdaS2 + (8*lambdaT3*Bar[lambdaT3])/3 + (8*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1
    ])/3 + lambdaH2Adj*(16*lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambdaS2Adj =
-2*g2^4 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*
    lambdaS2Adj^2

beta_lambdaT3GeneralQ000211I1 =
-12*g2^2*lambdaT3GeneralQ000211I1 - 6*gY^2*lambdaT3GeneralQ000211I1 + 2*lambdaS1*lambdaT3GeneralQ000211I1 + 8*
    lambdaS2*lambdaT3GeneralQ000211I1 + 4*lambdaS2Adj*lambdaT3GeneralQ000211I1 + lambda12*(8*
    lambdaT3GeneralQ000211I1 + 6*lambdaT3GeneralQ001300I1) + 4*lambdaH2*lambdaT3GeneralQ110200I1 + 24*
    lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000211I1] + 4*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1]

beta_lambdaT3GeneralQ000400I1 =
3*lambdaT3GeneralQ000211I1^2 + 12*lambdaS1*lambdaT3GeneralQ000400I1 + 9*lambdaT3GeneralQ001300I1^2 + 2*
    lambdaT3GeneralQ110200I1^2

beta_lambdaT3GeneralQ001300I1 =
6*lambda12*lambdaT3GeneralQ000211I1 + 18*lambdaS1*lambdaT3GeneralQ001300I1 + 4*lambdaH1*lambdaT3GeneralQ110200I1 +
    72*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ001300I1]

beta_lambdaT3GeneralQ021010I1 =
(-21*g2^2*lambdaT3GeneralQ021010I1)/2 - (9*gY^2*lambdaT3GeneralQ021010I1)/2 + 2*lambda12*lambdaT3GeneralQ021010I1 +
    2*lambdaH*lambdaT3GeneralQ021010I1 + 4*lambdaH1*lambdaT3GeneralQ021010I1 + 4*lambdaH2*lambdaT3GeneralQ021010I1
    + 4*lambdaH2Adj*lambdaT3GeneralQ021010I1 + 4*Bar[lambdaT3]*(Bar[lambdaT3GeneralQ000211I1] + 2*Bar[
    lambdaT3GeneralQ110200I1]) + lambdaT3GeneralQ021010I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
    6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110200I1 =
6*lambdaH2*lambdaT3GeneralQ000211I1 - (9*g2^2*lambdaT3GeneralQ110200I1)/2 - (3*gY^2*lambdaT3GeneralQ110200I1)/2 + 6*
    lambdaH*lambdaT3GeneralQ110200I1 + 2*lambdaS1*lambdaT3GeneralQ110200I1 + lambdaH1*(6*lambdaT3GeneralQ001300I1 +
     8*lambdaT3GeneralQ110200I1) + 12*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] + 24*lambdaT3GeneralQ000400I1*Bar
    [lambdaT3GeneralQ110200I1] + lambdaT3GeneralQ110200I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
    6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS1Sq =
6*lambdaT3GeneralQ001300I1*mS1Sq + 6*lambdaT3GeneralQ000211I1*mS2Sq + 2*lambdaS1*muT3GeneralS1Sq + 24*
    lambdaT3GeneralQ000400I1*Bar[muT3GeneralS1Sq]
```

2 A__-2

Additional quartics: 5; additional bilinears: 1; additional Yukawa: y2Conj
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

23 beta functions

```
beta_gY =
(43*gY^3)/6

beta_g2 =
(-5*g2^3)/2

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-6*gY^2*mS1Sq + 4*lambdaS1*mS1Sq + 6*lambda12*mS2Sq + 12*muT3GeneralS2Sq*Bar[lambdaT3GeneralQ001102I1] + 12*
    lambdaT3GeneralQ001102I1*Bar[muT3GeneralS2Sq]

beta_mS2Sq =
2*lambda12*mS1Sq - 12*g2^2*mS2Sq + (8*lambdaS2 + 4*lambdaS2Adj)*mS2Sq + 20*muT3GeneralS2Sq*Bar[
    lambdaT3GeneralQ000013I1] + 20*lambdaT3GeneralQ000013I1*Bar[muT3GeneralS2Sq]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 2*lambdaH1^2 + 6*lambdaH2
    ^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] +
    24*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
    Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
    Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
12*gY^4 + 6*lambda12^2 + 4*lambdaH1^2 - 12*gY^2*lambdaS1 + 10*lambdaS1^2 + 24*lambdaT3GeneralQ001102I1*Bar[
    lambdaT3GeneralQ001102I1]

beta_lambdaS2 =
22*g2^4 + 2*lambda12^2 + 4*lambdaH2^2 - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 8*lambdaS2*lambdaS2Adj + 16*lambdaS2Adj^2
     + (832*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1])/3 + (352*lambdaT3GeneralQ000013I1*Bar[
    lambdaT3GeneralQ000013I1])/3 + (8*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1])/3 + (16*
    lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3

beta_lambdaH1 =
3*gY^4 - (9*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambda12*lambdaH2 + 4*
    lambdaH1*lambdaS1 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ001102I1] + 12*
    lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 12*lambdaT3GeneralQ001102I1*Bar[
    lambdaT3GeneralQ110002I1] + lambdaH1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambdaH2 =
6*g2^4 + 2*lambda12*lambdaH1 - (33*g2^2*lambdaH2)/2 - (3*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 + 6*
    lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ020101I1*Bar[
    lambdaT3GeneralQ020101I1] + 20*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ110002I1] + 4*
    lambdaT3GeneralQ110002I1*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ110002I1]) + lambdaH2*(4*
    lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 4*lambda12*lambdaS1 + 8*lambda12*lambdaS2
     + 4*lambda12*lambdaS2Adj + 4*lambdaT3*Bar[lambdaT3] + 20*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ001102I1
```

```
    ] + 4*lambdaT3GeneralQ001102I1*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ001102I1]) + 4*
    lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ020101I1] + 8*lambdaT3GeneralQ110002I1*Bar[
    lambdaT3GeneralQ020101I1] + lambdaT3*(4*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
    6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
(-33*g2^2*lambdaH2Adj)/2 - (3*gY^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*
    lambdaS2 + (8*lambdaT3*Bar[lambdaT3])/3 - (8*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1])/3 +
    lambdaH2Adj*(16*lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu
    ]]])

beta_lambdaS2Adj =
-2*g2^4 + 3*lambdaH2Adj^2 - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2 - (224*
    lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1])/3 - (44*lambdaT3GeneralQ000013I1*Bar[
    lambdaT3GeneralQ000013I1])/3 - (4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1])/3 - (8*
    lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3

beta_lambdaT3GeneralQ000004I1 =
-24*g2^2*lambdaT3GeneralQ000004I1 + 12*lambdaS2*lambdaT3GeneralQ000004I1 - 8*lambdaS2Adj*lambdaT3GeneralQ000004I1 +
    11*lambdaT3GeneralQ000013I1^2 + lambdaT3GeneralQ001102I1^2 + 2*lambdaT3GeneralQ110002I1^2

beta_lambdaT3GeneralQ000013I1 =
-24*g2^2*lambdaT3GeneralQ000013I1 + 22*lambdaS2*lambdaT3GeneralQ000013I1 + 2*lambda12*lambdaT3GeneralQ001102I1 + 4*
    lambdaH2*lambdaT3GeneralQ110002I1 + 88*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000013I1]

beta_lambdaT3GeneralQ001102I1 =
-12*g2^2*lambdaT3GeneralQ001102I1 - 6*gY^2*lambdaT3GeneralQ001102I1 + 4*lambdaS1*lambdaT3GeneralQ001102I1 + 2*
    lambdaS2*lambdaT3GeneralQ001102I1 - 4*lambdaS2Adj*lambdaT3GeneralQ001102I1 + lambda12*(10*
    lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ001102I1) + 4*lambdaT3*lambdaT3GeneralQ020101I1 + 4*lambdaH1*
    lambdaT3GeneralQ110002I1 + 40*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ001102I1]

beta_lambdaT3GeneralQ020101I1 =
(-21*g2^2*lambdaT3GeneralQ020101I1)/2 - (9*gY^2*lambdaT3GeneralQ020101I1)/2 + 2*lambda12*lambdaT3GeneralQ020101I1 +
    2*lambdaH*lambdaT3GeneralQ020101I1 + 4*lambdaH1*lambdaT3GeneralQ020101I1 + 4*lambdaH2*lambdaT3GeneralQ020101I1
    - 4*lambdaH2Adj*lambdaT3GeneralQ020101I1 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3] + 8*
    lambdaT3GeneralQ110002I1*Bar[lambdaT3] + lambdaT3GeneralQ020101I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[
    Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110002I1 =
2*lambdaH1*lambdaT3GeneralQ001102I1 + 4*lambdaT3*lambdaT3GeneralQ020101I1 - (33*g2^2*lambdaT3GeneralQ110002I1)/2 -
    (3*gY^2*lambdaT3GeneralQ110002I1)/2 + 6*lambdaH*lambdaT3GeneralQ110002I1 + 2*lambdaS2*lambdaT3GeneralQ110002I1
    - 4*lambdaS2Adj*lambdaT3GeneralQ110002I1 + lambdaH2*(10*lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ110002I1)
    + 40*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ110002I1] + lambdaT3GeneralQ110002I1*(6*Tr[yd . Trans[Bar[yd
    ]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS2Sq =
2*lambdaT3GeneralQ001102I1*mS1Sq + 10*lambdaT3GeneralQ000013I1*mS2Sq - 12*g2^2*muT3GeneralS2Sq + 2*lambdaS2*
    muT3GeneralS2Sq - 4*lambdaS2Adj*muT3GeneralS2Sq + 40*lambdaT3GeneralQ000004I1*Bar[muT3GeneralS2Sq]
```

3 A_-4

Additional quartics: 0; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

17 beta functions

```
beta_gY =
(55*gY^3)/6

beta_g2 =
(-5*g2^3)/2

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-24*gY^2*mS1Sq + 4*lambdaS1*mS1Sq + 6*lambda12*mS2Sq

beta_mS2Sq =
2*lambda12*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq + (8*lambdaS2 + 4*lambdaS2Adj)*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 2*lambdaH1^2 + 6*lambdaH2
    ^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye
    ]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]]
     . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
192*gY^4 + 6*lambda12^2 + 4*lambdaH1^2 - 48*gY^2*lambdaS1 + 10*lambdaS1^2

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH2^2 + gY^2*(8*g2^2 - 12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 +
    8*lambdaS2*lambdaS2Adj + 16*lambdaS2Adj^2

beta_lambdaH1 =
12*gY^4 - (9*g2^2*lambdaH1)/2 - (51*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambda12*lambdaH2 + 4*
    lambdaH1*lambdaS1 + 12*lambdaT3*Bar[lambdaT3] + lambdaH1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]]
     + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + 2*lambda12*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*
    lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH2*(4*lambdaS2Adj + 6*Tr[
    yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
48*gY^4 - 12*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 4*lambda12*lambdaS1 + 8*
    lambda12*lambdaS2 + 4*lambda12*lambdaS2Adj + 4*lambdaT3*Bar[lambdaT3]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + lambdaT3*(4*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
     Trans[Bar[yu]]])

beta_lambdaH2Adj =
gY^2*(-4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*lambdaH2Adj + 2*
    lambdaH2Adj*lambdaS2 + (8*lambdaT3*Bar[lambdaT3])/3 + lambdaH2Adj*(16*lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] +
     2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
```

```
beta_lambdaS2Adj =
-2*g2^4 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*
     lambdaS2Adj^2
```

4 B_+1

Additional quartics: 1; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

19 beta functions

```
beta_gY =
(17*gY^3)/2

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 6*lambdaS1*mS1Sq + 4*lambda12*mS2Sq

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (27*gY^2*mS2Sq)/2 + 6*lambdaS2*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 + lambdaH1Adj
    ^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] + 8*lambdaT3GeneralQ022000I1*Bar[
    lambdaT3GeneralQ022000I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[
    Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]]
     - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1]

beta_lambdaS2 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((27*g2^2)/2 -
    27*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)
    /4 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] + 24*lambdaT3GeneralQ022000I1*Bar[
    lambdaT3GeneralQ022000I1] + lambdaH1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambdaH2 =
(9*g2^4)/4 + (27*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 15*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*
    lambdaH2^2 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] + lambdaH2*(6*Tr[yd . Trans[
    Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*
    lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*lambdaH2*
    lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
```

```
gY^2*(6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + lambda12Adj*
     lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 + 64*lambdaT3GeneralQ022000I1*Bar[
     lambdaT3GeneralQ022000I1] + lambdaH1Adj*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
     Trans[Bar[yu]]])

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(18*g2^2 - 15*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*
     lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] + lambdaH2Adj*(6*Tr[yd . Trans[Bar[yd]]] + 2*
     Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj + 2*
     lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3

beta_lambdaT3GeneralQ022000I1 =
-9*g2^2*lambdaT3GeneralQ022000I1 - 3*gY^2*lambdaT3GeneralQ022000I1 + 2*lambdaH*lambdaT3GeneralQ022000I1 + 8*lambdaH1
     *lambdaT3GeneralQ022000I1 + 4*lambdaH1Adj*lambdaT3GeneralQ022000I1 + 2*lambdaS1*lambdaT3GeneralQ022000I1 +
     lambdaT3GeneralQ022000I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
```

5 B__-1

Additional quartics: 7; additional bilinears: 1; additional Yukawa: none
Matching mass basis: Blocked
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

26 beta functions

```
beta_gY =
(43*gY^3)/6

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 6*lambdaS1*mS1Sq + 4*lambda12*mS2Sq + 6*muT3GeneralS1S2*Bar[
    lambdaS1S1bar2S2bar] + 6*lambdaS1S1bar2S2bar*Bar[muT3GeneralS1S2]

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + 6*lambdaS2*mS2Sq + 6*muT3GeneralS1S2*Bar[
    lambdaS1barS2S2bar2] + 6*lambdaS1barS2S2bar2*Bar[muT3GeneralS1S2]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 + lambdaH1Adj
    ^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 4*
    lambdaHHdagS1barS2barCross*Bar[lambdaHHdagS1barS2barCross] + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 4*
    lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ110101I1*Bar[lambdaT3GeneralQ110101I1] + lambdaH*(12*Tr[yd . Trans[
    Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[
    Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 8*
    lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 24*lambdaS1S1bar2S2bar*Bar[lambdaS1S1bar2S2bar]

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 8*lambdaS1bar2S2bar2*Bar
    [lambdaS1bar2S2bar2] + 24*lambdaS1barS2S2bar2*Bar[lambdaS1barS2S2bar2]

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)
    /4 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 24*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 6*
    lambdaT3*Bar[lambdaT3] + 3*lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1
    ]) + lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1S1bar2S2bar] + 2*Bar[
    lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] - 3*Bar[
    lambdaS1S1bar2S2bar] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans
    [Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2
    ^2 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 24*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 6*lambdaT3*Bar[
    lambdaT3] + 3*lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1]) +
    lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1barS2S2bar2] + 2*Bar[
    lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] - 3*Bar[
```

```
    lambdaS1barS2S2bar2] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans
    [Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambda12 - 3*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*
    lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 24*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 9*
    lambdaS1barS2S2bar2*(Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) + 9*lambdaS1S1bar2S2bar*(Bar[
    lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) + 3*lambdaT3*Bar[lambdaT3] + lambdaHHdagS1barS2barCross*(2*Bar
    [lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(Bar[
    lambdaHHdagS1barS2barCross] + 2*Bar[lambdaT3GeneralQ110101I1])

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 3*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*lambdaH2*
    lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + lambda12Adj*
    lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - 64*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 4*
    lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1S1bar2S2bar]) + (16*lambdaT3*Bar[
    lambdaT3])/3 + 4*lambdaT3GeneralQ110101I1*(Bar[lambdaS1S1bar2S2bar] - 2*Bar[lambdaT3GeneralQ110101I1]) + 4*
    lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) + lambdaH1Adj*(6*Tr[yd .
    Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*
    lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 64*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] - 4*lambdaHHdagS1barS2barCross
    *(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1barS2S2bar2]) + 16*lambdaT3*Bar[lambdaT3] - 4*
    lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) +
    lambdaT3GeneralQ110101I1*(-4*Bar[lambdaS1barS2S2bar2] + 8*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2Adj*(6*Tr[yd
     . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12Adj =
gY^2*(-6*g2^2 - 3*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj + 2*
    lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + 4*lambdaT3GeneralQ110101I1*Bar[lambdaHHdagS1barS2barCross] -
    64*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] - 4*lambdaS1barS2S2bar2*(5*Bar[lambdaS1barS2S2bar2] + Bar[
    lambdaS1S1bar2S2bar]) - 4*lambdaS1S1bar2S2bar*(Bar[lambdaS1barS2S2bar2] + 5*Bar[lambdaS1S1bar2S2bar]) + (4*
    lambdaT3*Bar[lambdaT3])/3 + 4*lambdaHHdagS1barS2barCross*Bar[lambdaT3GeneralQ110101I1]

beta_lambdaHHdagS2S2 =
-9*g2^2*lambdaHHdagS2S2 - 3*gY^2*lambdaHHdagS2S2 + 2*lambdaH*lambdaHHdagS2S2 + 8*lambdaH2*lambdaHHdagS2S2 + 4*
    lambdaH2Adj*lambdaHHdagS2S2 + 2*lambdaHHdagS2S2*lambdaS2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaS1bar2S2bar2] +
    lambdaHHdagS2S2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaHHdagS1barS1bar =
-9*g2^2*lambdaHHdagS1barS1bar - 3*gY^2*lambdaHHdagS1barS1bar + 2*lambdaH*lambdaHHdagS1barS1bar + 8*lambdaH1*
    lambdaHHdagS1barS1bar - 4*lambdaH1Adj*lambdaHHdagS1barS1bar + 2*lambdaHHdagS1barS1bar*lambdaS1 + 4*
    lambdaHHdagS2S2*lambdaS1bar2S2bar2 + lambdaHHdagS1barS1bar*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
    ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1bar2S2bar2 =
lambdaHHdagS1barS2barCross^2 - 9*g2^2*lambdaS1bar2S2bar2 - 3*gY^2*lambdaS1bar2S2bar2 + 8*lambda12*lambdaS1bar2S2bar2
     - 4*lambda12Adj*lambdaS1bar2S2bar2 + 2*lambdaS1*lambdaS1bar2S2bar2 + 5*lambdaS1barS2S2bar2^2 + 2*
    lambdaS1barS2S2bar2*lambdaS1S1bar2S2bar + 5*lambdaS1S1bar2S2bar^2 + 2*lambdaS1bar2S2bar2*lambdaS2 +
    lambdaT3GeneralQ110101I1^2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS2S2]

beta_lambdaS1barS2S2bar2 =
-9*g2^2*lambdaS1barS2S2bar2 - 3*gY^2*lambdaS1barS2S2bar2 + lambda12Adj*((-5*lambdaS1barS2S2bar2)/2 -
    lambdaS1S1bar2S2bar/2) + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar) + 12*lambdaS1barS2S2bar2*
    lambdaS2 + lambdaH2*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH2Adj*(-1/2*
    lambdaHHdagS1barS2barCross - lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(5*Bar[lambdaS1barS2S2bar2] +
    Bar[lambdaS1S1bar2S2bar])

beta_lambdaS1S1bar2S2bar =
lambda12Adj*(-1/2*lambdaS1barS2S2bar2 - (5*lambdaS1S1bar2S2bar)/2) - 9*g2^2*lambdaS1S1bar2S2bar - 3*gY^2*
    lambdaS1S1bar2S2bar + 12*lambdaS1*lambdaS1S1bar2S2bar + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar
    ) + lambdaH1*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH1Adj*(
    lambdaHHdagS1barS2barCross/2 + lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(Bar[lambdaS1barS2S2bar2] +
    5*Bar[lambdaS1S1bar2S2bar])

beta_lambdaHHdagS1barS2barCross =
-9*g2^2*lambdaHHdagS1barS2barCross - 3*gY^2*lambdaHHdagS1barS2barCross + 2*lambda12*lambdaHHdagS1barS2barCross +
    lambdaH2*(4*lambdaHHdagS1barS2barCross + 6*lambdaS1barS2S2bar2) + lambdaH1*(4*lambdaHHdagS1barS2barCross + 6*
    lambdaS1S1bar2S2bar) + lambdaH*(4*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH2Adj*(-2*
    lambdaHHdagS1barS2barCross - lambdaS1barS2S2bar2/2 - lambdaT3GeneralQ110101I1) + lambda12Adj*(-1/2*
    lambdaHHdagS1barS2barCross + lambdaT3GeneralQ110101I1) + lambdaH1Adj*(2*lambdaHHdagS1barS2barCross +
    lambdaS1S1bar2S2bar/2 + lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(8*Bar[lambdaHHdagS1barS2barCross] - 4*
    Bar[lambdaT3GeneralQ110101I1]) + lambdaHHdagS1barS2barCross*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
```

```
    ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110101I1 =
lambdaH1Adj*(-lambdaHHdagS1barS2barCross + lambdaS1S1bar2S2bar/2 - 2*lambdaT3GeneralQ110101I1) + lambda12Adj*(
    lambdaHHdagS1barS2barCross - lambdaT3GeneralQ110101I1/2) - 9*g2^2*lambdaT3GeneralQ110101I1 - 3*gY^2*
    lambdaT3GeneralQ110101I1 + 2*lambda12*lambdaT3GeneralQ110101I1 + lambdaH2Adj*(lambdaHHdagS1barS2barCross -
    lambdaS1barS2S2bar2/2 + 2*lambdaT3GeneralQ110101I1) + lambdaH*(-2*lambdaHHdagS1barS2barCross + 4*
    lambdaT3GeneralQ110101I1) + lambdaH2*(-6*lambdaS1barS2S2bar2 + 4*lambdaT3GeneralQ110101I1) + lambdaH1*(-6*
    lambdaS1S1bar2S2bar + 4*lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(-4*Bar[lambdaHHdagS1barS2barCross] + 8*
    Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
    ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS1S2 =
6*lambdaS1S1bar2S2bar*mS1Sq + 6*lambdaS1barS2S2bar2*mS2Sq - (9*g2^2*muT3GeneralS1S2)/2 - (3*gY^2*muT3GeneralS1S2)/2
    + 2*lambda12*muT3GeneralS1S2 - (3*lambda12Adj*muT3GeneralS1S2)/2 + 12*lambdaS1bar2S2bar2*Bar[muT3GeneralS1S2]
```

6 B_-3

Additional quartics: 1; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

19 beta functions

```
beta_gY =
(17*gY^3)/2

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (27*gY^2*mS1Sq)/2 + 6*lambdaS1*mS1Sq + 4*lambda12*mS2Sq

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + 6*lambdaS2*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 + lambdaH1Adj
    ^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] + 8*lambdaT3GeneralQ020002I1*Bar[
    lambdaT3GeneralQ020002I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[
    Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]]
     - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((27*g2^2)/2 -
    27*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1]

beta_lambdaH1 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambdaH1 - 15*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj
    ^2)/4 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] + lambdaH1*(6*Tr[yd . Trans[Bar[yd
    ]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2
    ^2 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] + 24*lambdaT3GeneralQ020002I1*Bar[
    lambdaT3GeneralQ020002I1] + lambdaH2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*
    lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*lambdaH2*
    lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
```

```
gY^2*(-18*g2^2 - 15*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + lambda12Adj
     *lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(6*Tr[yd . Trans[Bar[yd]]]
     + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(-6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*
     lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] - 64*lambdaT3GeneralQ020002I1*Bar[
     lambdaT3GeneralQ020002I1] + lambdaH2Adj*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
     Trans[Bar[yu]]])

beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj + 2*
     lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3

beta_lambdaT3GeneralQ020002I1 =
-9*g2^2*lambdaT3GeneralQ020002I1 - 3*gY^2*lambdaT3GeneralQ020002I1 + 2*lambdaH*lambdaT3GeneralQ020002I1 + 8*lambdaH2
     *lambdaT3GeneralQ020002I1 - 4*lambdaH2Adj*lambdaT3GeneralQ020002I1 + 2*lambdaS2*lambdaT3GeneralQ020002I1 +
     lambdaT3GeneralQ020002I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
```

7 C_+1

Additional quartics: 1; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

19 beta functions

```
beta_gY =
(17*gY^3)/2

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 6*lambdaS1*mS1Sq + 4*lambda12*mS2Sq

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (27*gY^2*mS2Sq)/2 + 6*lambdaS2*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 + lambdaH1Adj
    ^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] + 8*lambdaT3GeneralQ022000I1*Bar[
    lambdaT3GeneralQ022000I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[
    Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]]
     - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1]

beta_lambdaS2 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((27*g2^2)/2 -
    27*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)
    /4 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] + 24*lambdaT3GeneralQ022000I1*Bar[
    lambdaT3GeneralQ022000I1] + lambdaH1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambdaH2 =
(9*g2^4)/4 + (27*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 15*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*
    lambdaH2^2 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] + lambdaH2*(6*Tr[yd . Trans[
    Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*
    lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*lambdaH2*
    lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
```

```
gY^2*(6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + lambda12Adj*
     lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 + 64*lambdaT3GeneralQ022000I1*Bar[
     lambdaT3GeneralQ022000I1] + lambdaH1Adj*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
     Trans[Bar[yu]]])

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(18*g2^2 - 15*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*
     lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] + lambdaH2Adj*(6*Tr[yd . Trans[Bar[yd]]] + 2*
     Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj + 2*
     lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3

beta_lambdaT3GeneralQ022000I1 =
-9*g2^2*lambdaT3GeneralQ022000I1 - 3*gY^2*lambdaT3GeneralQ022000I1 + 2*lambdaH*lambdaT3GeneralQ022000I1 + 8*lambdaH1
     *lambdaT3GeneralQ022000I1 + 4*lambdaH1Adj*lambdaT3GeneralQ022000I1 + 2*lambdaS1*lambdaT3GeneralQ022000I1 +
     lambdaT3GeneralQ022000I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
```

8 C_-1

Additional quartics: 7; additional bilinears: 1; additional Yukawa: none
Matching mass basis: Blocked
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

26 beta functions

```
beta_gY =
(43*gY^3)/6

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 6*lambdaS1*mS1Sq + 4*lambda12*mS2Sq + 6*muT3GeneralS1S2*Bar[
    lambdaS1S1bar2S2bar] + 6*lambdaS1S1bar2S2bar*Bar[muT3GeneralS1S2]

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + 6*lambdaS2*mS2Sq + 6*muT3GeneralS1S2*Bar[
    lambdaS1barS2S2bar2] + 6*lambdaS1barS2S2bar2*Bar[muT3GeneralS1S2]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 + lambdaH1Adj
    ^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 4*
    lambdaHHdagS1barS2barCross*Bar[lambdaHHdagS1barS2barCross] + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 4*
    lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ110101I1*Bar[lambdaT3GeneralQ110101I1] + lambdaH*(12*Tr[yd . Trans[
    Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[
    Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 8*
    lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 24*lambdaS1S1bar2S2bar*Bar[lambdaS1S1bar2S2bar]

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 8*lambdaS1bar2S2bar2*Bar
    [lambdaS1bar2S2bar2] + 24*lambdaS1barS2S2bar2*Bar[lambdaS1barS2S2bar2]

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)
    /4 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 24*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 6*
    lambdaT3*Bar[lambdaT3] + 3*lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1
    ]) + lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1S1bar2S2bar] + 2*Bar[
    lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] - 3*Bar[
    lambdaS1S1bar2S2bar] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans
    [Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2
    ^2 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 24*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 6*lambdaT3*Bar[
    lambdaT3] + 3*lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1]) +
    lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1barS2S2bar2] + 2*Bar[
    lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] - 3*Bar[
```

```
    lambdaS1barS2S2bar2] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans
    [Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambda12 - 3*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*
    lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 24*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 9*
    lambdaS1barS2S2bar2*(Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) + 9*lambdaS1S1bar2S2bar*(Bar[
    lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) + 3*lambdaT3*Bar[lambdaT3] + lambdaHHdagS1barS2barCross*(2*Bar
    [lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(Bar[
    lambdaHHdagS1barS2barCross] + 2*Bar[lambdaT3GeneralQ110101I1])

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 3*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*lambdaH2*
    lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + lambda12Adj*
    lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - 64*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] + 4*
    lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1S1bar2S2bar]) + (16*lambdaT3*Bar[
    lambdaT3])/3 + 4*lambdaT3GeneralQ110101I1*(Bar[lambdaS1S1bar2S2bar] - 2*Bar[lambdaT3GeneralQ110101I1]) + 4*
    lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) + lambdaH1Adj*(6*Tr[yd .
    Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*
    lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 64*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] - 4*lambdaHHdagS1barS2barCross
    *(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1barS2S2bar2]) + 16*lambdaT3*Bar[lambdaT3] - 4*
    lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) +
    lambdaT3GeneralQ110101I1*(-4*Bar[lambdaS1barS2S2bar2] + 8*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2Adj*(6*Tr[yd
     . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12Adj =
gY^2*(-6*g2^2 - 3*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj + 2*
    lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + 4*lambdaT3GeneralQ110101I1*Bar[lambdaHHdagS1barS2barCross] -
    64*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] - 4*lambdaS1barS2S2bar2*(5*Bar[lambdaS1barS2S2bar2] + Bar[
    lambdaS1S1bar2S2bar]) - 4*lambdaS1S1bar2S2bar*(Bar[lambdaS1barS2S2bar2] + 5*Bar[lambdaS1S1bar2S2bar]) + (4*
    lambdaT3*Bar[lambdaT3])/3 + 4*lambdaHHdagS1barS2barCross*Bar[lambdaT3GeneralQ110101I1]

beta_lambdaHHdagS2S2 =
-9*g2^2*lambdaHHdagS2S2 - 3*gY^2*lambdaHHdagS2S2 + 2*lambdaH*lambdaHHdagS2S2 + 8*lambdaH2*lambdaHHdagS2S2 + 4*
    lambdaH2Adj*lambdaHHdagS2S2 + 2*lambdaHHdagS2S2*lambdaS2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaS1bar2S2bar2] +
    lambdaHHdagS2S2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaHHdagS1barS1bar =
-9*g2^2*lambdaHHdagS1barS1bar - 3*gY^2*lambdaHHdagS1barS1bar + 2*lambdaH*lambdaHHdagS1barS1bar + 8*lambdaH1*
    lambdaHHdagS1barS1bar - 4*lambdaH1Adj*lambdaHHdagS1barS1bar + 2*lambdaHHdagS1barS1bar*lambdaS1 + 4*
    lambdaHHdagS2S2*lambdaS1bar2S2bar2 + lambdaHHdagS1barS1bar*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
    ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1bar2S2bar2 =
lambdaHHdagS1barS2barCross^2 - 9*g2^2*lambdaS1bar2S2bar2 - 3*gY^2*lambdaS1bar2S2bar2 + 8*lambda12*lambdaS1bar2S2bar2
     - 4*lambda12Adj*lambdaS1bar2S2bar2 + 2*lambdaS1*lambdaS1bar2S2bar2 + 5*lambdaS1barS2S2bar2^2 + 2*
    lambdaS1barS2S2bar2*lambdaS1S1bar2S2bar + 5*lambdaS1S1bar2S2bar^2 + 2*lambdaS1bar2S2bar2*lambdaS2 +
    lambdaT3GeneralQ110101I1^2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS2S2]

beta_lambdaS1barS2S2bar2 =
-9*g2^2*lambdaS1barS2S2bar2 - 3*gY^2*lambdaS1barS2S2bar2 + lambda12Adj*((-5*lambdaS1barS2S2bar2)/2 -
    lambdaS1S1bar2S2bar/2) + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar) + 12*lambdaS1barS2S2bar2*
    lambdaS2 + lambdaH2*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH2Adj*(-1/2*
    lambdaHHdagS1barS2barCross - lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(5*Bar[lambdaS1barS2S2bar2] +
    Bar[lambdaS1S1bar2S2bar])

beta_lambdaS1S1bar2S2bar =
lambda12Adj*(-1/2*lambdaS1barS2S2bar2 - (5*lambdaS1S1bar2S2bar)/2) - 9*g2^2*lambdaS1S1bar2S2bar - 3*gY^2*
    lambdaS1S1bar2S2bar + 12*lambdaS1*lambdaS1S1bar2S2bar + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar
    ) + lambdaH1*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH1Adj*(
    lambdaHHdagS1barS2barCross/2 + lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(Bar[lambdaS1barS2S2bar2] +
    5*Bar[lambdaS1S1bar2S2bar])

beta_lambdaHHdagS1barS2barCross =
-9*g2^2*lambdaHHdagS1barS2barCross - 3*gY^2*lambdaHHdagS1barS2barCross + 2*lambda12*lambdaHHdagS1barS2barCross +
    lambdaH2*(4*lambdaHHdagS1barS2barCross + 6*lambdaS1barS2S2bar2) + lambdaH1*(4*lambdaHHdagS1barS2barCross + 6*
    lambdaS1S1bar2S2bar) + lambdaH*(4*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH2Adj*(-2*
    lambdaHHdagS1barS2barCross - lambdaS1barS2S2bar2/2 - lambdaT3GeneralQ110101I1) + lambda12Adj*(-1/2*
    lambdaHHdagS1barS2barCross + lambdaT3GeneralQ110101I1) + lambdaH1Adj*(2*lambdaHHdagS1barS2barCross +
    lambdaS1S1bar2S2bar/2 + lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(8*Bar[lambdaHHdagS1barS2barCross] - 4*
    Bar[lambdaT3GeneralQ110101I1]) + lambdaHHdagS1barS2barCross*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
```

```
    ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110101I1 =
lambdaH1Adj*(-lambdaHHdagS1barS2barCross + lambdaS1S1bar2S2bar/2 - 2*lambdaT3GeneralQ110101I1) + lambda12Adj*(
    lambdaHHdagS1barS2barCross - lambdaT3GeneralQ110101I1/2) - 9*g2^2*lambdaT3GeneralQ110101I1 - 3*gY^2*
    lambdaT3GeneralQ110101I1 + 2*lambda12*lambdaT3GeneralQ110101I1 + lambdaH2Adj*(lambdaHHdagS1barS2barCross -
    lambdaS1barS2S2bar2/2 + 2*lambdaT3GeneralQ110101I1) + lambdaH*(-2*lambdaHHdagS1barS2barCross + 4*
    lambdaT3GeneralQ110101I1) + lambdaH2*(-6*lambdaS1barS2S2bar2 + 4*lambdaT3GeneralQ110101I1) + lambdaH1*(-6*
    lambdaS1S1bar2S2bar + 4*lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(-4*Bar[lambdaHHdagS1barS2barCross] + 8*
    Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
    ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS1S2 =
6*lambdaS1S1bar2S2bar*mS1Sq + 6*lambdaS1barS2S2bar2*mS2Sq - (9*g2^2*muT3GeneralS1S2)/2 - (3*gY^2*muT3GeneralS1S2)/2
    + 2*lambda12*muT3GeneralS1S2 - (3*lambda12Adj*muT3GeneralS1S2)/2 + 12*lambdaS1bar2S2bar2*Bar[muT3GeneralS1S2]
```

9 C__-3

Additional quartics: 1; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

19 beta functions

```
beta_gY =
(17*gY^3)/2

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (27*gY^2*mS1Sq)/2 + 6*lambdaS1*mS1Sq + 4*lambda12*mS2Sq

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + 6*lambdaS2*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 + lambdaH1Adj
    ^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] + 8*lambdaT3GeneralQ020002I1*Bar[
    lambdaT3GeneralQ020002I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[
    Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]]
     - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((27*g2^2)/2 -
    27*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 - 3*
    lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1]

beta_lambdaH1 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambdaH1 - 15*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj
    ^2)/4 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] + lambdaH1*(6*Tr[yd . Trans[Bar[yd
    ]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2
    ^2 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] + 24*lambdaT3GeneralQ020002I1*Bar[
    lambdaT3GeneralQ020002I1] + lambdaH2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*
    lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*lambdaH2*
    lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
```

```
gY^2*(-18*g2^2 - 15*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + lambda12Adj
     *lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(6*Tr[yd . Trans[Bar[yd]]]
     + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(-6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*
     lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] - 64*lambdaT3GeneralQ020002I1*Bar[
     lambdaT3GeneralQ020002I1] + lambdaH2Adj*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
     Trans[Bar[yu]]])

beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj + 2*
     lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3

beta_lambdaT3GeneralQ020002I1 =
-9*g2^2*lambdaT3GeneralQ020002I1 - 3*gY^2*lambdaT3GeneralQ020002I1 + 2*lambdaH*lambdaT3GeneralQ020002I1 + 8*lambdaH2
     *lambdaT3GeneralQ020002I1 - 4*lambdaH2Adj*lambdaT3GeneralQ020002I1 + 2*lambdaS2*lambdaT3GeneralQ020002I1 +
     lambdaT3GeneralQ020002I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
```

10 D_+0

Additional quartics: 5; additional bilinears: 1; additional Yukawa: y1Conj
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

23 beta functions

```
beta_gY =
(43*gY^3)/6

beta_g2 =
(-5*g2^3)/2

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-12*g2^2*mS1Sq + (8*lambdaS1 + 4*lambdaS1Adj)*mS1Sq + 2*lambda12*mS2Sq + 20*muT3GeneralS1Sq*Bar[
    lambdaT3GeneralQ001300I1] + 20*lambdaT3GeneralQ001300I1*Bar[muT3GeneralS1Sq]

beta_mS2Sq =
6*lambda12*mS1Sq - 6*gY^2*mS2Sq + 4*lambdaS2*mS2Sq + 12*muT3GeneralS1Sq*Bar[lambdaT3GeneralQ000211I1] + 12*
    lambdaT3GeneralQ000211I1*Bar[muT3GeneralS1Sq]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 + 3*
    lambdaH1Adj^2 + 2*lambdaH2^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ021010I1*Bar[
    lambdaT3GeneralQ021010I1] + 24*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH*(12*Tr[yd .
    Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd .
    Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[
    yu]]]

beta_lambdaS1 =
22*g2^4 + 2*lambda12^2 + 4*lambdaH1^2 - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 8*lambdaS1*lambdaS1Adj + 16*lambdaS1Adj^2
     + (8*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1])/3 + (832*lambdaT3GeneralQ000400I1*Bar[
    lambdaT3GeneralQ000400I1])/3 + (352*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ001300I1])/3 + (16*
    lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3

beta_lambdaS2 =
12*gY^4 + 6*lambda12^2 + 4*lambdaH2^2 - 12*gY^2*lambdaS2 + 10*lambdaS2^2 + 24*lambdaT3GeneralQ000211I1*Bar[
    lambdaT3GeneralQ000211I1]

beta_lambdaH1 =
6*g2^4 - (33*g2^2*lambdaH1)/2 - (3*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 + 2*
    lambda12*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ021010I1*Bar[
    lambdaT3GeneralQ021010I1] + 20*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ110200I1] + 4*
    lambdaT3GeneralQ110200I1*(5*Bar[lambdaT3GeneralQ001300I1] + 4*Bar[lambdaT3GeneralQ110200I1]) + lambdaH1*(4*
    lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
3*gY^4 + 6*lambda12*lambdaH1 - (9*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 + 4*
    lambdaH2*lambdaS2 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ000211I1] + 12*
    lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 12*lambdaT3GeneralQ000211I1*Bar[
    lambdaT3GeneralQ110200I1] + lambdaH2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambda12 =
```

```
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 8*lambda12*lambdaS1 + 4*lambda12*
     lambdaS1Adj + 4*lambda12*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + 20*lambdaT3GeneralQ001300I1*Bar[
     lambdaT3GeneralQ000211I1] + 4*lambdaT3GeneralQ000211I1*(4*Bar[lambdaT3GeneralQ000211I1] + 5*Bar[
     lambdaT3GeneralQ001300I1]) + 4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
     lambdaH2*lambdaT3 + 4*Bar[lambdaT3GeneralQ021010I1]*(Bar[lambdaT3GeneralQ000211I1] + 2*Bar[
     lambdaT3GeneralQ110200I1]) + lambdaT3*(-4*lambdaH1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
      6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
(-33*g2^2*lambdaH1Adj)/2 - (3*gY^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + 2*lambdaH1Adj*
     lambdaS1 - (8*lambdaT3*Bar[lambdaT3])/3 + (8*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1])/3 +
     lambdaH1Adj*(16*lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu
     ]]])

beta_lambdaS1Adj =
-2*g2^4 + 3*lambdaH1Adj^2 - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2 - (4*
     lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1])/3 - (224*lambdaT3GeneralQ000400I1*Bar[
     lambdaT3GeneralQ000400I1])/3 - (44*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ001300I1])/3 - (8*
     lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3

beta_lambdaT3GeneralQ000211I1 =
-12*g2^2*lambdaT3GeneralQ000211I1 - 6*gY^2*lambdaT3GeneralQ000211I1 + 2*lambdaS1*lambdaT3GeneralQ000211I1 - 4*
     lambdaS1Adj*lambdaT3GeneralQ000211I1 + 4*lambdaS2*lambdaT3GeneralQ000211I1 + lambda12*(8*
     lambdaT3GeneralQ000211I1 + 10*lambdaT3GeneralQ001300I1) + 4*lambdaH2*lambdaT3GeneralQ110200I1 + 40*
     lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000211I1] + 4*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1]

beta_lambdaT3GeneralQ000400I1 =
lambdaT3GeneralQ000211I1^2 - 24*g2^2*lambdaT3GeneralQ000400I1 + 12*lambdaS1*lambdaT3GeneralQ000400I1 - 8*lambdaS1Adj
     *lambdaT3GeneralQ000400I1 + 11*lambdaT3GeneralQ001300I1^2 + 2*lambdaT3GeneralQ110200I1^2

beta_lambdaT3GeneralQ001300I1 =
2*lambda12*lambdaT3GeneralQ000211I1 - 24*g2^2*lambdaT3GeneralQ001300I1 + 22*lambdaS1*lambdaT3GeneralQ001300I1 + 4*
     lambdaH1*lambdaT3GeneralQ110200I1 + 88*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ001300I1]

beta_lambdaT3GeneralQ021010I1 =
(-21*g2^2*lambdaT3GeneralQ021010I1)/2 - (9*gY^2*lambdaT3GeneralQ021010I1)/2 + 2*lambda12*lambdaT3GeneralQ021010I1 +
     2*lambdaH*lambdaT3GeneralQ021010I1 + 4*lambdaH1*lambdaT3GeneralQ021010I1 + 4*lambdaH1Adj*
     lambdaT3GeneralQ021010I1 + 4*lambdaH2*lambdaT3GeneralQ021010I1 + 4*Bar[lambdaT3]*(Bar[lambdaT3GeneralQ000211I1]
      + 2*Bar[lambdaT3GeneralQ110200I1]) + lambdaT3GeneralQ021010I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar
     [ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110200I1 =
2*lambdaH2*lambdaT3GeneralQ000211I1 - (33*g2^2*lambdaT3GeneralQ110200I1)/2 - (3*gY^2*lambdaT3GeneralQ110200I1)/2 +
     6*lambdaH*lambdaT3GeneralQ110200I1 + 2*lambdaS1*lambdaT3GeneralQ110200I1 - 4*lambdaS1Adj*
     lambdaT3GeneralQ110200I1 + lambdaH1*(10*lambdaT3GeneralQ001300I1 + 8*lambdaT3GeneralQ110200I1) + 4*Bar[lambdaT3
     ]*Bar[lambdaT3GeneralQ021010I1] + 40*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ110200I1] +
     lambdaT3GeneralQ110200I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS1Sq =
10*lambdaT3GeneralQ001300I1*mS1Sq + 2*lambdaT3GeneralQ000211I1*mS2Sq - 12*g2^2*muT3GeneralS1Sq + 2*lambdaS1*
     muT3GeneralS1Sq - 4*lambdaS1Adj*muT3GeneralS1Sq + 40*lambdaT3GeneralQ000400I1*Bar[muT3GeneralS1Sq]
```

11 D_+2

Additional quartics: 0; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

17 beta functions

```
beta_gY =
(55*gY^3)/6

beta_g2 =
(-5*g2^3)/2

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + (8*lambdaS1 + 4*lambdaS1Adj)*mS1Sq + 2*lambda12*mS2Sq

beta_mS2Sq =
6*lambda12*mS1Sq - 24*gY^2*mS2Sq + 4*lambdaS2*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 + 3*
    lambdaH1Adj^2 + 2*lambdaH2^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans
    [Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[
    Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH1^2 + gY^2*(8*g2^2 - 12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 +
    8*lambdaS1*lambdaS1Adj + 16*lambdaS1Adj^2

beta_lambdaS2 =
192*gY^4 + 6*lambda12^2 + 4*lambdaH2^2 - 48*gY^2*lambdaS2 + 10*lambdaS2^2

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2
    + 2*lambda12*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + lambdaH1*(4*lambdaS1Adj + 6*Tr[yd .
    Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
12*gY^4 + 6*lambda12*lambdaH1 - (9*g2^2*lambdaH2)/2 - (51*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 + 4*
    lambdaH2*lambdaS2 + 12*lambdaT3*Bar[lambdaT3] + lambdaH2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]]
     + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
48*gY^4 - 12*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 8*lambda12*lambdaS1 + 4*
    lambda12*lambdaS1Adj + 4*lambda12*lambdaS2 + 4*lambdaT3*Bar[lambdaT3]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + lambdaT3*(-4*lambdaH1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu
    . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + 2*
    lambdaH1Adj*lambdaS1 - (8*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(16*lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] +
     2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
```

```
beta_lambdaS1Adj =
-2*g2^4 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*
     lambdaS1Adj^2
```

12 D_-2

Additional quartics: 5; additional bilinears: 1; additional Yukawa: y2Conj
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

23 beta functions

```
beta_gY =
(47*gY^3)/6

beta_g2 =
(-5*g2^3)/2

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + (8*lambdaS1 + 4*lambdaS1Adj)*mS1Sq + 2*lambda12*mS2Sq + 4*muT3GeneralS2Sq*Bar[
    lambdaT3GeneralQ001102I1] + 4*lambdaT3GeneralQ001102I1*Bar[muT3GeneralS2Sq]

beta_mS2Sq =
6*lambda12*mS1Sq + 4*lambdaS2*mS2Sq + 12*muT3GeneralS2Sq*Bar[lambdaT3GeneralQ000013I1] + 12*lambdaT3GeneralQ000013I1
    *Bar[muT3GeneralS2Sq]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 + 3*
    lambdaH1Adj^2 + 2*lambdaH2^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ020101I1*Bar[
    lambdaT3GeneralQ020101I1] + 8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH*(12*Tr[yd .
    Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd .
    Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[
    yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH1^2 + gY^2*(8*g2^2 - 12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 +
    8*lambdaS1*lambdaS1Adj + 16*lambdaS1Adj^2 + 8*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1]

beta_lambdaS2 =
6*lambda12^2 + 4*lambdaH2^2 + 10*lambdaS2^2 + 288*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1] + 108*
    lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ000013I1] + 12*lambdaT3GeneralQ001102I1*Bar[
    lambdaT3GeneralQ001102I1] + 8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1]

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2
    + 2*lambda12*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ110002I1*Bar[
    lambdaT3GeneralQ001102I1] + 4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 4*
    lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH1*(4*lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] +
    2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
6*lambda12*lambdaH1 - (9*g2^2*lambdaH2)/2 - (3*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 + 4*lambdaH2*
    lambdaS2 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 12*
    lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ110002I1] + 4*lambdaT3GeneralQ110002I1*(3*Bar[
    lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ110002I1]) + lambdaH2*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 8*lambda12*lambdaS1 + 4*lambda12*
    lambdaS1Adj + 4*lambda12*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ000013I1*Bar[
```

```
    lambdaT3GeneralQ001102I1] + 4*lambdaT3GeneralQ001102I1*(3*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[
    lambdaT3GeneralQ001102I1]) + 4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ020101I1] + 8*lambdaT3GeneralQ110002I1*Bar[
    lambdaT3GeneralQ020101I1] + lambdaT3*(-4*lambdaH1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
    6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + 2*
    lambdaH1Adj*lambdaS1 - (8*lambdaT3*Bar[lambdaT3])/3 - (8*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1
    ])/3 + lambdaH1Adj*(16*lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
    Bar[yu]]])

beta_lambdaS1Adj =
-2*g2^4 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*
    lambdaS1Adj^2

beta_lambdaT3GeneralQ000004I1 =
12*lambdaS2*lambdaT3GeneralQ000004I1 + 9*lambdaT3GeneralQ000013I1^2 + 3*lambdaT3GeneralQ001102I1^2 + 2*
    lambdaT3GeneralQ110002I1^2

beta_lambdaT3GeneralQ000013I1 =
18*lambdaS2*lambdaT3GeneralQ000013I1 + 6*lambda12*lambdaT3GeneralQ001102I1 + 4*lambdaH2*lambdaT3GeneralQ110002I1 +
    72*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000013I1]

beta_lambdaT3GeneralQ001102I1 =
-12*g2^2*lambdaT3GeneralQ001102I1 - 6*gY^2*lambdaT3GeneralQ001102I1 + 8*lambdaS1*lambdaT3GeneralQ001102I1 + 4*
    lambdaS1Adj*lambdaT3GeneralQ001102I1 + 2*lambdaS2*lambdaT3GeneralQ001102I1 + lambda12*(6*
    lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ001102I1) + 4*lambdaT3*lambdaT3GeneralQ020101I1 + 4*lambdaH1*
    lambdaT3GeneralQ110002I1 + 24*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ001102I1]

beta_lambdaT3GeneralQ020101I1 =
(-21*g2^2*lambdaT3GeneralQ020101I1)/2 - (9*gY^2*lambdaT3GeneralQ020101I1)/2 + 2*lambda12*lambdaT3GeneralQ020101I1 +
    2*lambdaH*lambdaT3GeneralQ020101I1 + 4*lambdaH1*lambdaT3GeneralQ020101I1 - 4*lambdaH1Adj*
    lambdaT3GeneralQ020101I1 + 4*lambdaH2*lambdaT3GeneralQ020101I1 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3] + 8*
    lambdaT3GeneralQ110002I1*Bar[lambdaT3] + lambdaT3GeneralQ020101I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[
    Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110002I1 =
6*lambdaH1*lambdaT3GeneralQ001102I1 + 12*lambdaT3*lambdaT3GeneralQ020101I1 - (9*g2^2*lambdaT3GeneralQ110002I1)/2 -
    (3*gY^2*lambdaT3GeneralQ110002I1)/2 + 6*lambdaH*lambdaT3GeneralQ110002I1 + 2*lambdaS2*lambdaT3GeneralQ110002I1
    + lambdaH2*(6*lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ110002I1) + 24*lambdaT3GeneralQ000004I1*Bar[
    lambdaT3GeneralQ110002I1] + lambdaT3GeneralQ110002I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
    6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS2Sq =
6*lambdaT3GeneralQ001102I1*mS1Sq + 6*lambdaT3GeneralQ000013I1*mS2Sq + 2*lambdaS2*muT3GeneralS2Sq + 24*
    lambdaT3GeneralQ000004I1*Bar[muT3GeneralS2Sq]
```

13 E_+0

Additional quartics: 6; additional bilinears: 1; additional Yukawa: y1Conj
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

28 beta functions

```
beta_gY =
(47*gY^3)/6

beta_g2 =
(-11*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-12*g2^2*mS1Sq + (8*lambdaS1 + 4*lambdaS1Adj)*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq + 20*muT3GeneralS1Sq*Bar[
    lambdaT3GeneralQ001300I1] + 20*lambdaT3GeneralQ001300I1*Bar[muT3GeneralS1Sq]

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq + (8*lambdaS2 + 4*lambdaS2Adj)*mS2Sq + 4*
    muT3GeneralS1Sq*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) + 12*lambdaT3GeneralQ000211I1
    *Bar[muT3GeneralS1Sq] + 4*lambdaT3GeneralQ000211I2*Bar[muT3GeneralS1Sq]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 + 3*
    lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ021010I1*Bar[
    lambdaT3GeneralQ021010I1] + 24*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH*(12*Tr[yd .
    Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd .
    Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[
    yu]]]

beta_lambdaS1 =
22*g2^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 - 24*g2^2*lambdaS1 + 14*
    lambdaS1^2 + 8*lambdaS1*lambdaS1Adj + 16*lambdaS1Adj^2 + (8*lambdaT3GeneralQ000211I1*(3*Bar[
    lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]))/3 + lambdaT3GeneralQ000211I2*((8*Bar[
    lambdaT3GeneralQ000211I1])/3 + 4*Bar[lambdaT3GeneralQ000211I2]) + (832*lambdaT3GeneralQ000400I1*Bar[
    lambdaT3GeneralQ000400I1])/3 + (352*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ001300I1])/3 + (16*
    lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 + gY^2*(8*g2^2 -
     12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 8*lambdaS2*lambdaS2Adj + 16*lambdaS2Adj^2 + 8*
    lambdaT3GeneralQ000211I1*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) +
    lambdaT3GeneralQ000211I2*(8*Bar[lambdaT3GeneralQ000211I1] + (20*Bar[lambdaT3GeneralQ000211I2])/3)

beta_lambdaH1 =
6*g2^4 - (33*g2^2*lambdaH1)/2 - (3*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 + (6*
    lambda12 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + 4*
    lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 20*lambdaT3GeneralQ001300I1*Bar[
    lambdaT3GeneralQ110200I1] + 4*lambdaT3GeneralQ110200I1*(5*Bar[lambdaT3GeneralQ001300I1] + 4*Bar[
    lambdaT3GeneralQ110200I1]) + lambdaH1*(4*lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
    6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*
    lambdaH2 + 4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + 4*
```

```
     lambdaT3GeneralQ110200I1*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) + 4*
     lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 12*lambdaT3GeneralQ000211I1*Bar[
     lambdaT3GeneralQ110200I1] + 4*lambdaT3GeneralQ000211I2*Bar[lambdaT3GeneralQ110200I1] + lambdaH2*(4*lambdaS2Adj
     + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(63*g2^4)/5 - 24*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 + 4*lambdaH1*
     lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 + 4*
     lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + lambda12*(4*lambdaS1Adj + 4*lambdaS2Adj) + 2*lambdaT3*
     Bar[lambdaT3] + 4*lambdaT3GeneralQ001300I1*(5*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) +
     4*lambdaT3GeneralQ000211I2*(Bar[lambdaT3GeneralQ000211I2] + Bar[lambdaT3GeneralQ001300I1]) + 4*
     lambdaT3GeneralQ000211I1*(4*Bar[lambdaT3GeneralQ000211I1] + 5*Bar[lambdaT3GeneralQ001300I1]) + 2*
     lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1]

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
     lambdaH2*lambdaT3 + 2*Bar[lambdaT3GeneralQ021010I1]*(2*Bar[lambdaT3GeneralQ000211I1] - Bar[
     lambdaT3GeneralQ000211I2] + 4*Bar[lambdaT3GeneralQ110200I1]) + lambdaT3*((-4*lambda12Adj)/3 - 2*lambda12Cross -
      2*lambdaH1Adj - (4*lambdaH2Adj)/3 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[
     Bar[yu]]])

beta_lambdaH1Adj =
(-33*g2^2*lambdaH1Adj)/2 - (3*gY^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + 12*lambda12Adj*
     lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 - (8*lambdaT3GeneralQ021010I1*Bar[
     lambdaT3GeneralQ021010I1])/9 + lambdaH1Adj*(16*lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
     ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj + gY^2*(4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj +
     8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 - (8*lambdaT3GeneralQ021010I1*
     Bar[lambdaT3GeneralQ021010I1])/9 + lambdaH2Adj*(16*lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[
     Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj
      + 20*lambdaS1Adj^2 - (4*lambdaT3GeneralQ000211I2*Bar[lambdaT3GeneralQ000211I1])/3 - (4*
     lambdaT3GeneralQ000211I1*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]))/3 - (224*
     lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000400I1])/3 - (44*lambdaT3GeneralQ001300I1*Bar[
     lambdaT3GeneralQ001300I1])/3 - (8*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3

beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) - 24*g2^2*
     lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2 - (4*lambdaT3GeneralQ000211I2*Bar[
     lambdaT3GeneralQ000211I2])/3

beta_lambda12Adj =
-24*g2^2*lambda12Adj - 6*gY^2*lambda12Adj + 8*lambda12*lambda12Adj + (16*lambda12Adj*lambda12Cross)/3 + 3*
     lambdaH1Adj*lambdaH2Adj + 2*lambda12Adj*lambdaS1 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*
     lambda12Adj*lambdaS2Adj - (2*lambdaT3*Bar[lambdaT3])/9 - (lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1
     ])/3

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 6*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 + 2*
     lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
      lambdaT3*Bar[lambdaT3] + 8*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I2] + 4*lambdaT3GeneralQ001300I1
     *Bar[lambdaT3GeneralQ000211I2] + lambdaT3GeneralQ000211I2*(8*Bar[lambdaT3GeneralQ000211I1] + 10*Bar[
     lambdaT3GeneralQ000211I2] + 4*Bar[lambdaT3GeneralQ001300I1]) - lambdaT3GeneralQ021010I1*Bar[
     lambdaT3GeneralQ021010I1]

beta_lambdaT3GeneralQ000211I1 =
-24*g2^2*lambdaT3GeneralQ000211I1 - 6*gY^2*lambdaT3GeneralQ000211I1 + 2*lambdaS1*lambdaT3GeneralQ000211I1 +
     lambdaS1Adj*(-4*lambdaT3GeneralQ000211I1 - 2*lambdaT3GeneralQ000211I2) + lambdaS2Adj*(4*
     lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2) + lambdaS2*(8*lambdaT3GeneralQ000211I1 + 2*
     lambdaT3GeneralQ000211I2) + lambda12Cross*(4*lambdaT3GeneralQ000211I2 + 4*lambdaT3GeneralQ001300I1) + lambda12
     *(8*lambdaT3GeneralQ000211I1 + 10*lambdaT3GeneralQ001300I1) + 4*lambdaH2*lambdaT3GeneralQ110200I1 + 8*
     lambdaT3GeneralQ000400I1*(5*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) + 2*Bar[lambdaT3]*
     Bar[lambdaT3GeneralQ021010I1]

beta_lambdaT3GeneralQ000211I2 =
-24*g2^2*lambdaT3GeneralQ000211I2 - 6*gY^2*lambdaT3GeneralQ000211I2 + 8*lambda12*lambdaT3GeneralQ000211I2 + 2*
     lambdaS1*lambdaT3GeneralQ000211I2 + 2*lambdaS1Adj*lambdaT3GeneralQ000211I2 + 2*lambdaS2*
     lambdaT3GeneralQ000211I2 - 2*lambdaS2Adj*lambdaT3GeneralQ000211I2 + lambda12Cross*(16*lambdaT3GeneralQ000211I1
     + 20*lambdaT3GeneralQ000211I2 + 8*lambdaT3GeneralQ001300I1) + 16*lambdaT3GeneralQ000400I1*Bar[
     lambdaT3GeneralQ000211I2] - 2*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1]

beta_lambdaT3GeneralQ000400I1 =
3*lambdaT3GeneralQ000211I1^2 + 2*lambdaT3GeneralQ000211I1*lambdaT3GeneralQ000211I2 + lambdaT3GeneralQ000211I2^2 -
     24*g2^2*lambdaT3GeneralQ000400I1 + 12*lambdaS1*lambdaT3GeneralQ000400I1 - 8*lambdaS1Adj*
```

```
    lambdaT3GeneralQ000400I1 + 11*lambdaT3GeneralQ001300I1^2 + 2*lambdaT3GeneralQ110200I1^2

beta_lambdaT3GeneralQ001300I1 =
lambda12*(6*lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2) + lambda12Cross*(4*lambdaT3GeneralQ000211I1 + 4*
    lambdaT3GeneralQ000211I2) - 24*g2^2*lambdaT3GeneralQ001300I1 + 22*lambdaS1*lambdaT3GeneralQ001300I1 + 4*
    lambdaH1*lambdaT3GeneralQ110200I1 + 88*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ001300I1]

beta_lambdaT3GeneralQ021010I1 =
(-33*g2^2*lambdaT3GeneralQ021010I1)/2 - (9*gY^2*lambdaT3GeneralQ021010I1)/2 + 2*lambda12*lambdaT3GeneralQ021010I1 -
    2*lambda12Adj*lambdaT3GeneralQ021010I1 - 2*lambda12Cross*lambdaT3GeneralQ021010I1 + 2*lambdaH*
    lambdaT3GeneralQ021010I1 + 4*lambdaH1*lambdaT3GeneralQ021010I1 - (4*lambdaH1Adj*lambdaT3GeneralQ021010I1)/3 +
    4*lambdaH2*lambdaT3GeneralQ021010I1 - (4*lambdaH2Adj*lambdaT3GeneralQ021010I1)/3 + 2*Bar[lambdaT3]*(2*Bar[
    lambdaT3GeneralQ000211I1] - Bar[lambdaT3GeneralQ000211I2] + 4*Bar[lambdaT3GeneralQ110200I1]) +
    lambdaT3GeneralQ021010I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110200I1 =
lambdaH2*(6*lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2) - (33*g2^2*lambdaT3GeneralQ110200I1)/2 - (3*gY^2*
    lambdaT3GeneralQ110200I1)/2 + 6*lambdaH*lambdaT3GeneralQ110200I1 + 2*lambdaS1*lambdaT3GeneralQ110200I1 - 4*
    lambdaS1Adj*lambdaT3GeneralQ110200I1 + lambdaH1*(10*lambdaT3GeneralQ001300I1 + 8*lambdaT3GeneralQ110200I1) + 4*
    Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] + 40*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ110200I1] +
    lambdaT3GeneralQ110200I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS1Sq =
10*lambdaT3GeneralQ001300I1*mS1Sq + (6*lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2)*mS2Sq - 12*g2^2*
    muT3GeneralS1Sq + 2*lambdaS1*muT3GeneralS1Sq - 4*lambdaS1Adj*muT3GeneralS1Sq + 40*lambdaT3GeneralQ000400I1*Bar[
    muT3GeneralS1Sq]
```

14 E_+2

Additional quartics: 0; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

21 beta functions

```
beta_gY =
(71*gY^3)/6

beta_g2 =
(-11*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + (8*lambdaS1 + 4*lambdaS1Adj)*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq - 24*gY^2*mS2Sq + (8*lambdaS2 + 4*lambdaS2Adj)*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 + 3*
    lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]]
     + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] -
     4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 + gY^2*(8*g2^2 -
     12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 8*lambdaS1*lambdaS1Adj + 16*lambdaS1Adj^2

beta_lambdaS2 =
22*g2^4 + 192*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 + gY^2*(32*g2^2
     - 48*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 8*lambdaS2*lambdaS2Adj + 16*lambdaS2Adj^2

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2
    + (6*lambda12 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + lambdaH1*(4*
    lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
6*g2^4 + 12*gY^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (51*gY^2*lambdaH2)/2 + 6*lambdaH
    *lambdaH2 + 4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH2*(4*
    lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(63*g2^4)/5 + 48*gY^4 - 24*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 +
    4*lambdaH1*lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 + 4*
    lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + lambda12*(4*lambdaS1Adj + 4*lambdaS2Adj) + 2*lambdaT3*
    Bar[lambdaT3]

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + lambdaT3*((-4*lambda12Adj)/3 - 2*lambda12Cross - 2*lambdaH1Adj - (4*lambdaH2Adj)/3 + 6*Tr[
    yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
```

```
gY^2*(4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + 12*
     lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(16*lambdaS1Adj +
      6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj + gY^2*(8*g2^2 - (51*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj +
     8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 + lambdaH2Adj*(16*lambdaS2Adj +
     6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) - 24*g2^2*
     lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2

beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 + gY^2*(32*g2^2 - 48*lambdaS2Adj) - 24*g2^2*
     lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2

beta_lambda12Adj =
gY^2*(16*g2^2 - 30*lambda12Adj) - 24*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + (16*lambda12Adj*lambda12Cross)/3 +
     3*lambdaH1Adj*lambdaH2Adj + 2*lambda12Adj*lambdaS1 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*
     lambda12Adj*lambdaS2Adj - (2*lambdaT3*Bar[lambdaT3])/9

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 30*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 + 2*
     lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
      lambdaT3*Bar[lambdaT3]
```

15 E__-2

Additional quartics: 6; additional bilinears: 1; additional Yukawa: y2Conj
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

28 beta functions

```
beta_gY =
(47*gY^3)/6

beta_g2 =
(-11*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + (8*lambdaS1 + 4*lambdaS1Adj)*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq + 4*
    muT3GeneralS2Sq*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) + 12*lambdaT3GeneralQ001102I1
    *Bar[muT3GeneralS2Sq] + 4*lambdaT3GeneralQ001102I2*Bar[muT3GeneralS2Sq]

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq + (8*lambdaS2 + 4*lambdaS2Adj)*mS2Sq + 20*muT3GeneralS2Sq*Bar[
    lambdaT3GeneralQ000013I1] + 20*lambdaT3GeneralQ000013I1*Bar[muT3GeneralS2Sq]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 + 3*
    lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ020101I1*Bar[
    lambdaT3GeneralQ020101I1] + 24*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH*(12*Tr[yd .
    Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd .
    Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[
    yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 + gY^2*(8*g2^2 -
     12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 8*lambdaS1*lambdaS1Adj + 16*lambdaS1Adj^2 + 8*
    lambdaT3GeneralQ001102I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) +
    lambdaT3GeneralQ001102I2*(8*Bar[lambdaT3GeneralQ001102I1] + (20*Bar[lambdaT3GeneralQ001102I2])/3)

beta_lambdaS2 =
22*g2^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 - 24*g2^2*lambdaS2 + 14*
    lambdaS2^2 + 8*lambdaS2*lambdaS2Adj + 16*lambdaS2Adj^2 + (832*lambdaT3GeneralQ000004I1*Bar[
    lambdaT3GeneralQ000004I1])/3 + (352*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ000013I1])/3 + (8*
    lambdaT3GeneralQ001102I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]))/3 +
    lambdaT3GeneralQ001102I2*((8*Bar[lambdaT3GeneralQ001102I1])/3 + 4*Bar[lambdaT3GeneralQ001102I2]) + (16*
    lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2
    + (6*lambda12 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + 4*
    lambdaT3GeneralQ110002I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) + 4*
    lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 12*lambdaT3GeneralQ001102I1*Bar[
    lambdaT3GeneralQ110002I1] + 4*lambdaT3GeneralQ001102I2*Bar[lambdaT3GeneralQ110002I1] + lambdaH1*(4*lambdaS1Adj
    + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
6*g2^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (3*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 +
     4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ020101I1*
```

```
    Bar[lambdaT3GeneralQ020101I1] + 20*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ110002I1] + 4*
    lambdaT3GeneralQ110002I1*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ110002I1]) + lambdaH2*(4*
    lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(63*g2^4)/5 - 24*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 + 4*lambdaH1*
    lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 + 4*
    lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + lambda12*(4*lambdaS1Adj + 4*lambdaS2Adj) + 2*lambdaT3*
    Bar[lambdaT3] + 4*lambdaT3GeneralQ001102I1*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ001102I1])
    + 4*lambdaT3GeneralQ001102I2*(Bar[lambdaT3GeneralQ000013I1] + Bar[lambdaT3GeneralQ001102I2]) + 4*
    lambdaT3GeneralQ000013I1*(5*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) + 2*
    lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1]

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ020101I1] - 2*lambdaT3GeneralQ001102I2*Bar[
    lambdaT3GeneralQ020101I1] + 8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ020101I1] + lambdaT3*((-4*
    lambda12Adj)/3 - 2*lambda12Cross - 2*lambdaH1Adj - (4*lambdaH2Adj)/3 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
    Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + 12*
    lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 - (4*lambdaT3GeneralQ020101I1*
    Bar[lambdaT3GeneralQ020101I1])/3 + lambdaH1Adj*(16*lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[
    Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj - (33*g2^2*lambdaH2Adj)/2 - (3*gY^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*
    lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 - (4*lambdaT3GeneralQ020101I1*Bar[
    lambdaT3GeneralQ020101I1])/3 + lambdaH2Adj*(16*lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye
    ]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) - 24*g2^2*
    lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2 - (4*lambdaT3GeneralQ001102I2*Bar[
    lambdaT3GeneralQ001102I2])/3

beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj
     + 20*lambdaS2Adj^2 - (224*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1])/3 - (44*
    lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ000013I1])/3 - (4*lambdaT3GeneralQ001102I2*Bar[
    lambdaT3GeneralQ001102I1])/3 - (4*lambdaT3GeneralQ001102I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[
    lambdaT3GeneralQ001102I2]))/3 - (8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3

beta_lambda12Adj =
-24*g2^2*lambda12Adj - 6*gY^2*lambda12Adj + 8*lambda12*lambda12Adj + (16*lambda12Adj*lambda12Cross)/3 + 3*
    lambdaH1Adj*lambdaH2Adj + 2*lambda12Adj*lambdaS1 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*
    lambda12Adj*lambdaS2Adj - (2*lambdaT3*Bar[lambdaT3])/9 - (lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1
    ])/3

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 6*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 + 2*
    lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
     lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ001102I2] + 8*lambdaT3GeneralQ001102I1
    *Bar[lambdaT3GeneralQ001102I2] + lambdaT3GeneralQ001102I2*(4*Bar[lambdaT3GeneralQ000013I1] + 8*Bar[
    lambdaT3GeneralQ001102I1] + 10*Bar[lambdaT3GeneralQ001102I2]) - lambdaT3GeneralQ020101I1*Bar[
    lambdaT3GeneralQ020101I1]

beta_lambdaT3GeneralQ000004I1 =
-24*g2^2*lambdaT3GeneralQ000004I1 + 12*lambdaS2*lambdaT3GeneralQ000004I1 - 8*lambdaS2Adj*lambdaT3GeneralQ000004I1 +
    11*lambdaT3GeneralQ000013I1^2 + 3*lambdaT3GeneralQ001102I1^2 + 2*lambdaT3GeneralQ001102I1*
    lambdaT3GeneralQ001102I2 + lambdaT3GeneralQ001102I2^2 + 2*lambdaT3GeneralQ110002I1^2

beta_lambdaT3GeneralQ000013I1 =
-24*g2^2*lambdaT3GeneralQ000013I1 + 22*lambdaS2*lambdaT3GeneralQ000013I1 + lambda12*(6*lambdaT3GeneralQ001102I1 + 2*
    lambdaT3GeneralQ001102I2) + lambda12Cross*(4*lambdaT3GeneralQ001102I1 + 4*lambdaT3GeneralQ001102I2) + 4*
    lambdaH2*lambdaT3GeneralQ110002I1 + 88*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000013I1]

beta_lambdaT3GeneralQ001102I1 =
-24*g2^2*lambdaT3GeneralQ001102I1 - 6*gY^2*lambdaT3GeneralQ001102I1 + 2*lambdaS2*lambdaT3GeneralQ001102I1 + lambda12
    *(10*lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ001102I1) + lambdaS2Adj*(-4*lambdaT3GeneralQ001102I1 - 2*
    lambdaT3GeneralQ001102I2) + lambdaS1Adj*(4*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2) + lambdaS1
    *(8*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2) + lambda12Cross*(4*lambdaT3GeneralQ000013I1 + 4*
    lambdaT3GeneralQ001102I2) + 2*lambdaT3*lambdaT3GeneralQ020101I1 + 4*lambdaH1*lambdaT3GeneralQ110002I1 + 8*
    lambdaT3GeneralQ000004I1*(5*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2])

beta_lambdaT3GeneralQ001102I2 =
```

```
-24*g2^2*lambdaT3GeneralQ001102I2 - 6*gY^2*lambdaT3GeneralQ001102I2 + 8*lambda12*lambdaT3GeneralQ001102I2 + 2*
     lambdaS1*lambdaT3GeneralQ001102I2 - 2*lambdaS1Adj*lambdaT3GeneralQ001102I2 + 2*lambdaS2*
     lambdaT3GeneralQ001102I2 + 2*lambdaS2Adj*lambdaT3GeneralQ001102I2 + lambda12Cross*(8*lambdaT3GeneralQ000013I1 +
      16*lambdaT3GeneralQ001102I1 + 20*lambdaT3GeneralQ001102I2) - 2*lambdaT3*lambdaT3GeneralQ020101I1 + 16*
     lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ001102I2]

beta_lambdaT3GeneralQ020101I1 =
(-33*g2^2*lambdaT3GeneralQ020101I1)/2 - (9*gY^2*lambdaT3GeneralQ020101I1)/2 + 2*lambda12*lambdaT3GeneralQ020101I1 -
     2*lambda12Adj*lambdaT3GeneralQ020101I1 - 2*lambda12Cross*lambdaT3GeneralQ020101I1 + 2*lambdaH*
     lambdaT3GeneralQ020101I1 + 4*lambdaH1*lambdaT3GeneralQ020101I1 - 2*lambdaH1Adj*lambdaT3GeneralQ020101I1 + 4*
     lambdaH2*lambdaT3GeneralQ020101I1 - 2*lambdaH2Adj*lambdaT3GeneralQ020101I1 + 4*lambdaT3GeneralQ001102I1*Bar[
     lambdaT3] - 2*lambdaT3GeneralQ001102I2*Bar[lambdaT3] + 8*lambdaT3GeneralQ110002I1*Bar[lambdaT3] +
     lambdaT3GeneralQ020101I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110002I1 =
lambdaH1*(6*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2) + 4*lambdaT3*lambdaT3GeneralQ020101I1 - (33*g2^2*
     lambdaT3GeneralQ110002I1)/2 - (3*gY^2*lambdaT3GeneralQ110002I1)/2 + 6*lambdaH*lambdaT3GeneralQ110002I1 + 2*
     lambdaS2*lambdaT3GeneralQ110002I1 - 4*lambdaS2Adj*lambdaT3GeneralQ110002I1 + lambdaH2*(10*
     lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ110002I1) + 40*lambdaT3GeneralQ000004I1*Bar[
     lambdaT3GeneralQ110002I1] + lambdaT3GeneralQ110002I1*(6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
     6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS2Sq =
(6*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2)*mS1Sq + 10*lambdaT3GeneralQ000013I1*mS2Sq - 12*g2^2*
     muT3GeneralS2Sq + 2*lambdaS2*muT3GeneralS2Sq - 4*lambdaS2Adj*muT3GeneralS2Sq + 40*lambdaT3GeneralQ000004I1*Bar[
     muT3GeneralS2Sq]
```

16 E__-4

Additional quartics: 0; additional bilinears: 0; additional Yukawa: none
Matching mass basis: CheckedNotMatched
General-theory matching and physical Matchete–RGBeta normalisation are not certified.

21 beta functions

```
beta_gY =
(71*gY^3)/6

beta_g2 =
(-11*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i], gen[$j
    ]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
    gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
     gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd . Trans[Bar[yd]]]*
    Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu . Trans[Bar[yu
    ]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_mS1Sq =
-12*g2^2*mS1Sq - 24*gY^2*mS1Sq + (8*lambdaS1 + 4*lambdaS1Adj)*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq + (8*lambdaS2 + 4*lambdaS2Adj)*mS2Sq

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 + 3*
    lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]]
     + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] -
     4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 192*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 + gY^2*(32*g2^2
     - 48*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 8*lambdaS1*lambdaS1Adj + 16*lambdaS1Adj^2

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 + gY^2*(8*g2^2 -
     12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 8*lambdaS2*lambdaS2Adj + 16*lambdaS2Adj^2

beta_lambdaH1 =
6*g2^4 + 12*gY^4 - (33*g2^2*lambdaH1)/2 - (51*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2
     + (6*lambda12 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + lambdaH1*(4*
    lambdaS1Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*
    lambdaH2 + 4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH2*(4*
    lambdaS2Adj + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambda12 =
(63*g2^4)/5 + 48*gY^4 - 24*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 +
    4*lambdaH1*lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 + 4*
    lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + lambda12*(4*lambdaS1Adj + 4*lambdaS2Adj) + 2*lambdaT3*
    Bar[lambdaT3]

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 + 4*
    lambdaH2*lambdaT3 + lambdaT3*((-4*lambda12Adj)/3 - 2*lambda12Cross - 2*lambdaH1Adj - (4*lambdaH2Adj)/3 + 6*Tr[
    yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
```

```
gY^2*(-8*g2^2 - (51*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj + 12*
     lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(16*lambdaS1Adj +
      6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj + gY^2*(-4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj +
      8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 + lambdaH2Adj*(16*lambdaS2Adj +
      6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 + gY^2*(32*g2^2 - 48*lambdaS1Adj) - 24*g2^2*
     lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2

beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) - 24*g2^2*
     lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2

beta_lambda12Adj =
gY^2*(16*g2^2 - 30*lambda12Adj) - 24*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + (16*lambda12Adj*lambda12Cross)/3 +
     3*lambdaH1Adj*lambdaH2Adj + 2*lambda12Adj*lambdaS1 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*
     lambda12Adj*lambdaS2Adj - (2*lambdaT3*Bar[lambdaT3])/9

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 30*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 + 2*
     lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
      lambdaT3*Bar[lambdaT3]
```